Airbus Ship Detection Using Machine Learning and Deep Learning Approaches

Detecting ships in satellite imagery is a significant challenge in maritime surveillance, environmental monitoring, and navigation safety. The Airbus Ship Detection dataset frames this as a binary classification problem: determining whether an image contains a ship (has_ship = 1) or not (has_ship = 0). This task is complicated by a pronounced class imbalance, the small size of ships relative to the image, varying lighting conditions, cloud cover, and background clutter across large oceanic scenes.

Approach and Model Implementations

1. Traditional Machine Learning Approach

A lightweight and interpretable machine learning pipeline was developed, focusing on handcrafted features to avoid the computational expense of deep learning. Each image was resized to 256×256 pixels, normalized, and described using Hu Moments (for shape), Haralick Features (for texture), and Color Histograms (for color content).
Five classic models—LDA, KNN, Naïve Bayes, SVM, and Random Forest—were trained and evaluated. Random Forest (n_estimators=140, max_depth=30) achieved 92.48% accuracy on a sample of 15,000 images, maintaining the dataset's natural imbalance. This method proved efficient and interpretable, providing a reliable baseline for comparison.

2. Simple CNN (Custom Model)

A custom Convolutional Neural Network was constructed to automatically learn spatial and texture-based features. The network architecture comprised three convolutional blocks (Conv2D → ReLU → MaxPool), followed by fully connected layers and a sigmoid classifier.
The model was trained on a balanced subset of 70,000 images (35,000 ship, 35,000 no-ship), with extensive data augmentation (rotations and flips) to improve robustness. Using the Adam optimizer (learning rate 1e-4) and binary cross-entropy loss, the model reached a validation accuracy of 93.97% after 30 epochs, demonstrating stable convergence and strong generalization across diverse ship appearances.

3. Transfer Learning CNN (DenseNet169)

DenseNet169, pretrained on ImageNet, was fine-tuned for the ship detection task. The convolutional base was frozen, and a custom classification head (BatchNorm → Dropout(0.5) → Global Max Pooling → Dense(1664, ReLU) → Dense(128, ReLU) → Dense(1, Sigmoid) was trained on 10,000 images, preserving the original class imbalance.
With the Adam optimizer (learning rate 1e-4) and early stopping, the model achieved 93.9% validation accuracy, using an adjusted decision threshold of 0.6 to balance precision and recall. Transfer learning offered strong accuracy and stability with relatively little data and training time.

Problems and Challenges

Dataset-Level Issues:

The dataset exhibited a severe class imbalance, with the majority of images containing no ships, which led to a tendency toward negative predictions. Many images included clouds, haze, or water reflections, introducing noise that could obscure ships. Ships were often small relative to the image, making global feature-based detection difficult. The high resolution (768×768) and large dataset size (~190,000 images) also necessitated careful sampling and resource management to fit within GPU constraints.

Model-Specific Issues:

- Traditional ML approaches were sensitive to feature selection and had limited representational power, making generalization across diverse image conditions difficult. Combining features improved performance but could introduce redundancy.
- The Simple CNN was prone to overfitting on limited data; strong augmentation and regularization were needed for model stability. Misclassifications often occurred in cloudy or coastal scenes.
- For transfer learning, pretrained ImageNet filters sometimes did not fully adapt to the characteristics of satellite imagery, and managing GPU memory was crucial during fine-tuning. Adjusting the decision threshold was necessary to obtain balanced precision and recall, given the class imbalance.

Conclusion

All three methods were effective for binary classification of satellite images in the Airbus Ship Detection dataset. The Simple CNN achieved the highest validation accuracy (93.97%), closely followed by DenseNet169 (93.9%), while Random Forest (92.48%) provided a fast and interpretable alternative. This study demonstrates that both handcrafted and deep learning approaches can yield strong results for maritime image analysis. The findings underscore the trade-offs between interpretability, complexity, and performance, and provide a solid foundation for future work in ship segmentation and localization.